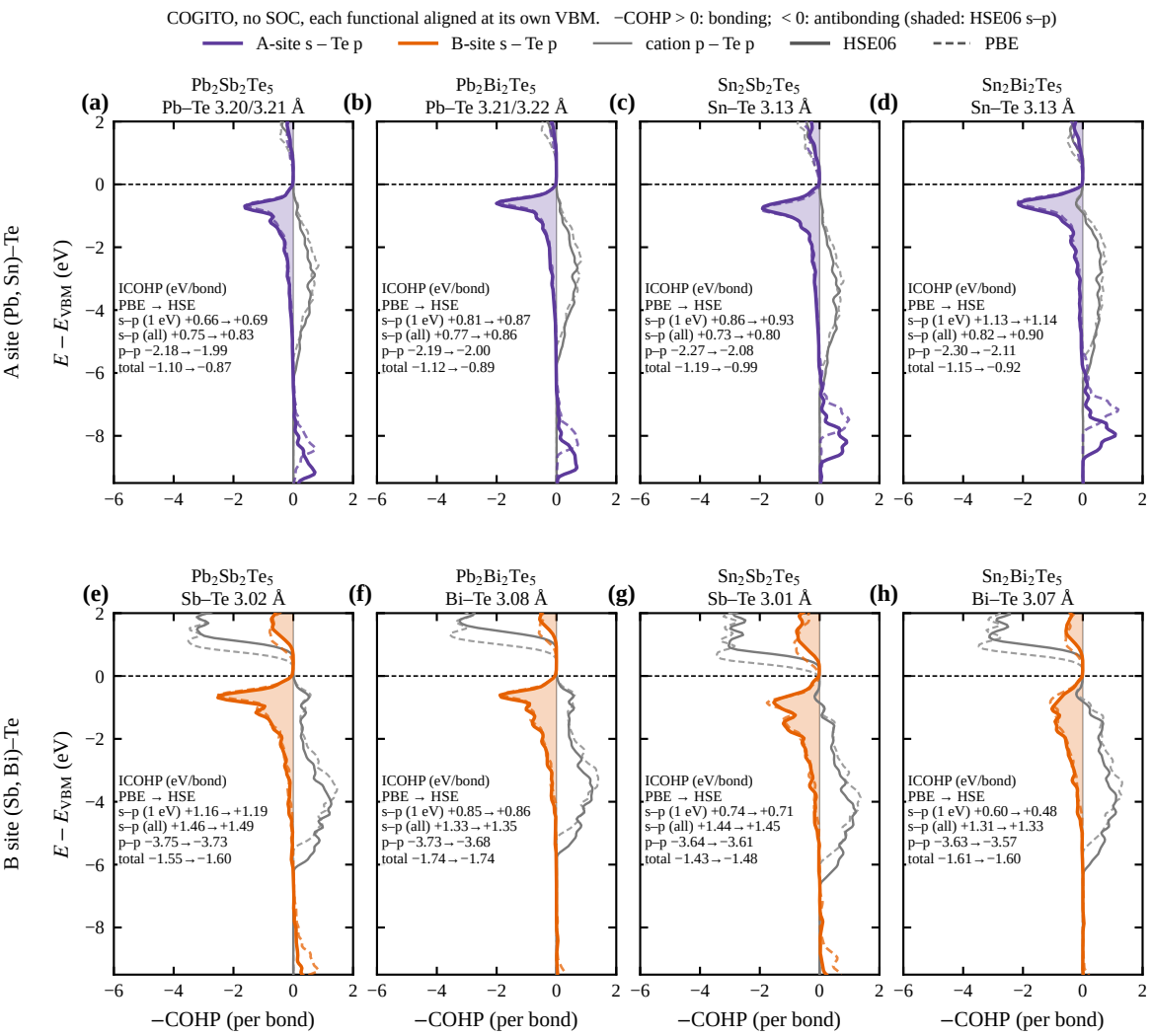

COGITO, no SOC, each functional aligned at its own VBM. −COHP > 0: bonding; < 0: antibonding (shaded: HSE06 s–p)
A-site s – Te p
B-site s – Te p
cation p – Te p
HSE06
PBE
(a)
Pb2Sb2Te5
Pb–Te 3.20/3.21 Å
(b)
Pb2Bi2Te5
Pb–Te 3.21/3.22 Å
(c)
Sn2Sb2Te5
Sn–Te 3.13 Å
(d)
Sn2Bi2Te5
Sn–Te 3.13 Å
A site (Pb, Sn)–Te
B site (Sb, Bi)–Te
E − EVBM (eV)
ICOHP (eV/bond)
PBE → HSE
s–p (1 eV) +0.66 → +0.69
s–p (all) +0.75 → +0.83
p–p −2.18 → −1.99
total −1.10 → −0.87
s–p (1 eV) +0.81 → +0.87
s–p (all) +0.77 → +0.86
p–p −2.19 → −2.00
total −1.12 → −0.89
s–p (1 eV) +0.86 → +0.93
s–p (all) +0.73 → +0.80
p–p −2.27 → −2.08
total −1.19 → −0.99
s–p (1 eV) +1.13 → +1.14
s–p (all) +0.82 → +0.90
p–p −2.30 → −2.11
total −1.15 → −0.92
(e)
Pb2Sb2Te5
Sb–Te 3.02 Å
(f)
Pb2Bi2Te5
Bi–Te 3.08 Å
(g)
Sn2Sb2Te5
Sb–Te 3.01 Å
(h)
Sn2Bi2Te5
Bi–Te 3.07 Å
s–p (1 eV) +1.16 → +1.19
s–p (all) +1.46 → +1.49
p–p −3.75 → −3.73
total −1.55 → −1.60
s–p (1 eV) +0.85 → +0.86
s–p (all) +1.33 → +1.35
p–p −3.73 → −3.68
total −1.74 → −1.74
s–p (1 eV) +0.74 → +0.71
s–p (all) +1.44 → +1.45
p–p −3.64 → −3.61
total −1.43 → −1.48
s–p (1 eV) +0.60 → +0.48
s–p (all) +1.31 → +1.33
p–p −3.63 → −3.57
total −1.61 → −1.60
−COHP (per bond)